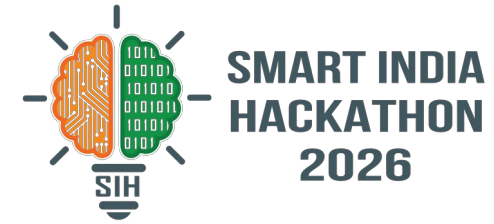


SMART INDIA HACKATHON 2026

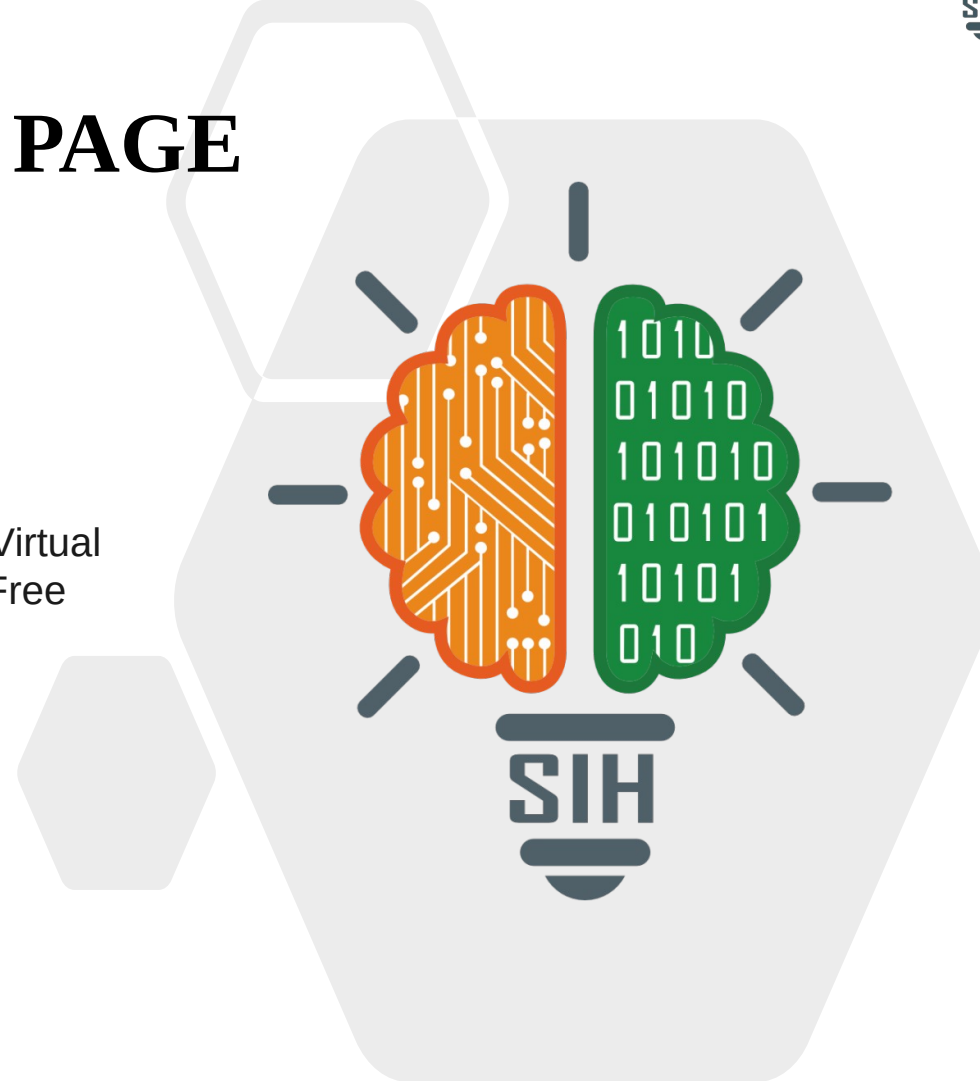


TITLE PAGE

ANVESHA

Belief-Driven Coarse PAT for Mobile FSOC Terminals

- **Problem Statement ID** – SIH26169
- **Problem Statement Title** – Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals
- **Theme** – Smart Automation / Space Technology
- **PS Category** – Software
- **Team ID** – _____
- **Team Name (Registered on portal)** – Regnum Carya



Where to look · how wide to look · what not to chase

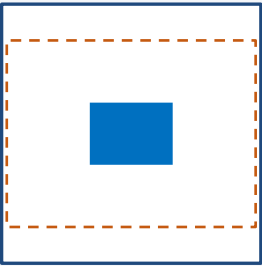
IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

ANVESHA: a probability map of the beacon decides **where to point** and **how wide to look**.

Measured (SIL, 48 runs each): 87 % of runs acquired ≤ 2 s
vs 12–15 % for spiral baselines (slide 4)

How it addresses the problem

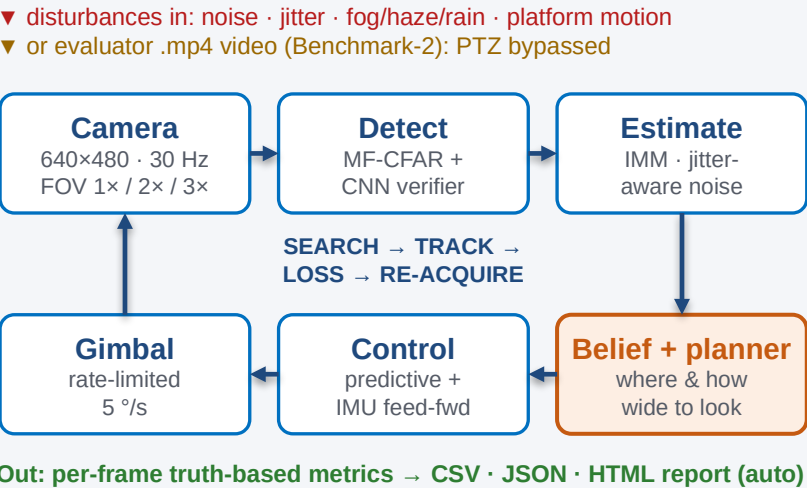


4°×3° FOV = **7.7 %** of the screen
5 °/s gimbal $\Rightarrow \leq 10^\circ$ slew in 2 s
Blind raster needs $\approx 40\text{--}50^\circ$ of slew
 $\Rightarrow$ **~8–10 s**
Measured spiral baselines A / B:
median **7.8 / 9.2 s**
PS asks ≤ 2 s from a random start

To scale: screen 12.5°×12.5° · ■ 4°×3° base FOV · ▭ 12°×9° wide FOV

Detecting a bright spot is easy. Knowing **where to look** is the real problem.

Detailed explanation — the closed loop



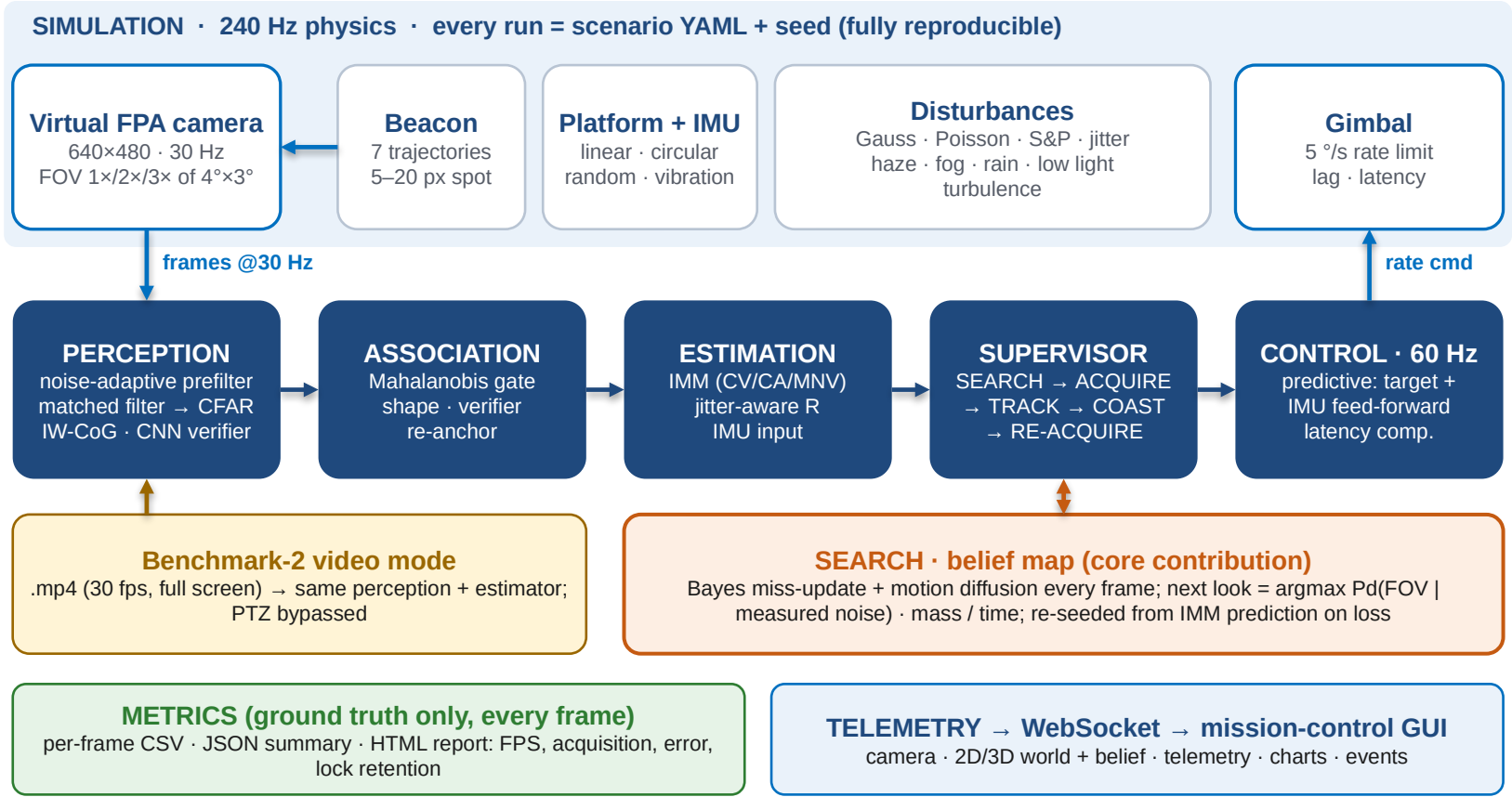
Innovation and uniqueness

- 1 Belief-driven search**
every empty frame is evidence: probability is removed where the beacon was not seen; next look = best P(detect) per second
- 2 Disturbance-aware FOV**
FOV (1×/2×/3×) chosen by predicted detection probability from the measured noise: go wide only when the beacon stays detectable
- 3 One belief, whole lifecycle**
on loss the search restarts from the tracker's prediction; jitter-aware estimator stops the gimbal chasing jitter

Typical public approach	Limitation	ANVESHA approach
Fixed spiral / raster at base FOV	Open loop; ignores empty frames; ~8–10 s worst case	Recursive Bayesian belief + next-best-look planner
Wide-FOV mode switched by a fixed rule	SNR loss of a wider FOV is not modelled	FOV picked from a measured-noise detection model
Re-acquire = restart spiral / Kalman coast	Throws away what the tracker knew	Belief re-seeded from IMM predicted mean & covariance

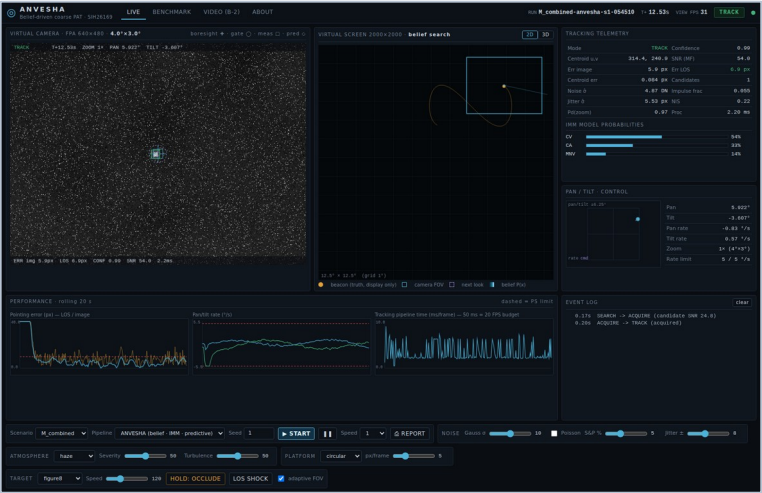
Based on our review of 134 papers and 20 public SIH26169 repositories, we did not identify a coarse-PAT system that combines these three elements.

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



Multi-rate loop: 240 Hz physics · 60 Hz control · 30 Hz camera + perception. Prior art we build on, not claimed as novel: MF-CFAR, IMM, CNN verifier, IMU feed-forward, predictive control.
LOS = line of sight · MF-CFAR = matched filter + constant-false-alarm-rate threshold · IW-CoG = intensity-weighted centroid · IMM = interacting multiple model (CV / CA / manoeuvre) · R = measurement-noise covariance

Working prototype (live GUI)



Scenario *M_combined* (haze + S&P + jitter + platform), TRACK mode

❖ Technologies to be used

Technology	Why
Python 3.11 · NumPy · SciPy · OpenCV	vectorised ROI processing on a CPU
PyTorch → ONNX Runtime	6.8 k-param CNN verifier, CPU inference
FastAPI + WebSocket	one engine feeds GUI, bench & reports
HTML/JS + three.js	2D/3D mission-control view, no build step
PyInstaller	standalone Windows .exe (planned)

FEASIBILITY AND VIABILITY

❖ Analysis of the feasibility of the idea

Measured in simulation (software-in-the-loop), 3 seeds · 16 scenarios · same seeds for all pipelines · baseline PID gains grid-tuned · laptop CPU only (no GPU used)

ANVESHA · scenario	Acq. ≤2 s	LOS err	Image err	Loss*	Re-acq	Loop FPS
A Clean sky	3/3 · 1.3 s	1.5	1.5	0 %	—	167
B Gaussian $\sigma=20$	3/3 · 1.2 s	1.9	1.9	0 %	—	166
C Poisson	3/3 · 1.1 s	0.8	0.8	0 %	—	55
D Salt & pepper 10 %	3/3 · 1.1 s	6.7	6.7	0 %	—	142
E Jitter ± 10 px	3/3 · 1.3 s	4.6	8.9	0 %	—	170
F Platform linear	3/3 · 1.1 s	4.4	4.4	0 %	—	167
F2 Vibration 7/13 Hz	2/3 · 1.4 s	20.5	20.5	0 %	—	168
G Fog	3/3 · 1.3 s	1.5	1.5	0 %	—	136
H Haze	3/3 · 1.2 s	3.5	3.5	0 %	—	136
I Low light	3/3 · 1.2 s	2.2	2.2	0 %	—	54
J Fast manoeuvre	3/3 · 1.3 s	18.6	18.6	0 %	—	168
K Occlusion	3/3 · 1.3 s	16.2	16.2	1 %	0.10 s	147
L Rain	3/3 · 1.4 s	1.9	1.9	0 %	—	124
M Combined	3/3 · 1.2 s	4.9	7.8	0 %	—	122
N All PS maxima	0/3	21.0	23.6	99 %	n/a (1X)	51
O Look-alike distractors	1/3 · 2.9 s	1.5	1.5	0 %	—	170

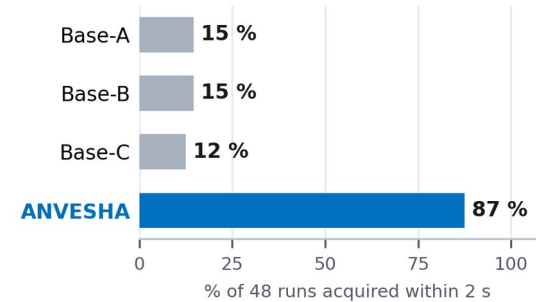
All PS checks met: **ANVESHA 11/16 scenarios** · baselines A/B/C 0/0/0

Limits: acq ≤ 2 s · error ≤ 10 px · loss < 5 % · re-acq ≤ 1 s · ≥ 20 FPS. *excl. scripted occlusion · LOS = excl. camera jitter

Measured gaps — shown, not hidden

- **All PS maxima at once (N):** acquired in 33 % of runs; look-alikes (O) acquired in 2.9 s (> 2 s) — rejecting static look-alikes costs time.
- **LOS error 16–21 px** (> 10) under vibration, fast manoeuvre, occlusion.
- **Jitter floor: ± 20 px** white jitter alone = 16.3 px RMS (analytic) — no coarse gimbal can meet 10 px image error. At ± 10 px (E): image 8.9 vs LOS 4.6 px.

Same seeds, all pipelines



Ablation — remove one part (3 seeds):

- **belief** → **spiral search**: runs ≤ 2 s 83 % → 7 %
- **adaptive FOV off**: runs ≤ 2 s 83 % → 20 %
- **jitter-aware R off (E)**: LOS 4.6 → 19.3 px
- **IMU feed-fwd off (F)**: LOS 4.4 → 5.8 px
- **CNN verifier off**: false-lock frames 614 → 667 (sum)

Re-acquisition (K, 2 occlusions): mean 0.10 s, max 0.27 s, 0 of 6 unrecovered

Benchmark-2 video rehearsal: centroid RMSE 0.13 px over 300 frames of 2000×2000 @30 fps; 88 FPS incl. decoding

Median acquisition when acquired: **ANVESHA 1.3 s** · baselines 6.4–9.2 s

Potential challenges and risks	Strategies for overcoming these challenges
Sim-to-real gap	video mode already accepts real footage; tracker needs only frames, encoder / IMU and rate commands → HIL
± 20 px jitter floor	report LOS vs image error; jitter-aware R; fine stage (FSM) hand-off
Faint beacon, clutter, look-alikes	joint position × detectability belief; CNN veto + shape tests; static look-alike rejection; beacon modulation ID (next)
CPU budget	ROI processing; CNN only on candidate patches; FPS logged each frame
CNN domain shift	CNN only verifies; classical MF-CFAR stays primary

Implementation status

Implemented

sim, 16 scenarios, 4 pipelines, belief search, IMM, GUI, video mode, reports

Prototype

CNN verifier: recall 90 %, FPR 23 % on held-out synthetic patches

Planned

.exe build · HIL rig · real footage

❖ Potential impact on the target audience

1

ISRO / DoS PAT teams

Design and regression-test coarse-PAT logic for mobile ground terminals, UAV/HAP relays and LEO-to-ground links before hardware exists.

2

Academia & students

Open, seeded, repeatable PAT testbed for courses and research — no optics lab, camera or gimbal needed.

3

Indian FSOC industry / start-ups

The tracker exchanges only frames, encoder / IMU readings and rate commands — a real camera + pan-tilt can replace the simulator (HIL).

❖ Benefits of the solution (social, economic, environmental, etc.)

Benefit	What it means
Economic	Removes the cost barrier named in the PS: iterate without cameras, pan-tilt units, beacons or an optical bench; hardware only for final validation.
Technical	Every run = (scenario, pipeline, seed): disturbance sweeps that cannot be reproduced outdoors, with truth-based logs.
Strategic	Indigenous, auditable coarse-PAT software stack for Indian optical-communication missions.
Social / skills	Hands-on PAT, computer-vision and control learning for students across India.
Environmental	Fewer vehicle / field trials needed for early tuning.

Why acquisition matters: an on-orbit optical link demonstration reported a 22 s acquisition time [Wang et al., Opt. Lett. 2023].

Roadmap: software-in-the-loop → hardware-in-the-loop (same tracker, simulator swapped for hardware)

1 · SIL (now)

simulator, 16 scenarios, benchmarks, GUI

2 · Video / real footage

Benchmark-2 mode done; real recordings next

3 · HIL rig

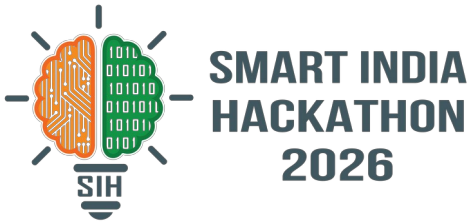
COTS camera + pan-tilt + LED/laser beacon

4 · Fine-stage hand-off

FSM / quadrant detector removes residual jitter



RESEARCH AND REFERENCES



❖ Details / Links of the reference and research work

134	papers in a structured literature database (PAT, FOU scanning, centroiding, IMM, gimbal control)	20	public SIH26169 repositories analysed (README level) for differentiation	16×4×3	scenarios × pipelines × seeds on identical seeds, plus a one-part-removed ablation study
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[9] Li L. et al., “Noise adaptive fading Kalman filter for free-space laser communication beacon tracking,” Appl. Opt., 2016. doi.org/10.1364/ao.55.008486

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Based on our review of 134 papers and 20 public SIH26169 repositories, we did not identify a coarse-PAT system that combines a recursively updated belief map, FOV selection from measured noise, and belief-seeded re-acquisition.

PS coverage

10 mandatory functions implemented · 5 performance specs measured per scenario (all met in 11/16) · auto performance logs · Benchmark-2 video mode